Valuations, Simplex Embeddings, and Product Rules in Finite Quantum Logics

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This paper compares three classicality tests for finite quantum logics: incidence-level tests using valuations, colorings, and partition-logical representations; operational tests using simplex embeddings of probability tables; and operator-functional tests using spectral and product rules for observable values. Two simplex audits are reported. The first is a restricted projector-cone benchmark using the listed rank-one projectors as preparations and effects. The second closes the rational vector-generated scenarios under unit effects, residual outcomes of projection-valued measures, complements, coarse grainings, and optionally normalized coarse-grained preparations. Exact rational projector-cone rows have primal and dual certificates; closure rows are numerical unless separately certified.

I. INTRODUCTION

In Hilbert spaces of dimension at least three, quantum propositions cannot in general be assigned pre-existing, context-independent truth values compatible with the functional structure of sharp measurements [1–9]. In quantum-logical language this is first a statement about pasted Boolean algebras, orthomodular posets, or orthogonality hypergraphs. At the operational level it becomes a question about whether a specified prepare-and-measure table admits a classical explanation [10–15]. Generalized-noncontextual ontological models [16], the simplex-embeddability criterion of Schmid, Selby, Wolfe, Kunjwal, and Spekkens [17], and the linear program of Selby, Wolfe, Schmid, Sainz, and Rossi [18] give one precise way to formulate that operational question. Operator-valued parity proofs, such as Greenberger–Horne–Zeilinger (GHZ) and Mermin arguments [19–21], retain still another part of the same Hilbert-space description: algebraic relations among observables.

The aim of this paper is to keep these retained structures visible. Incidence data lead to valuation, coloring, and partition-logic questions [6, 22, 23]. Convex-operational data lead to simplex-embedding questions for specified preparations, effects, probabilities, and operational equivalences. Operator data lead to functional-calculus and product-rule questions for assigned observable values. These are related tests, but they are not a hierarchy. The same physical construction can change its classicality status when contexts, effects, local factors, coarse grainings, or operational equivalences are added or forgotten.

This organization complements existing classifications rather than replacing them. The graph-theoretic approach of Cabello, Severini, and Winter relates exclusivity graphs to invariants such as independence numbers and the Lovász theta function [24]. The hypergraph approach of Acín, Fritz, Leverrier, and Sainz formulates contextuality scenarios directly as compatibility/exclusivity structures with probabilistic models [25]. The sheaf-theoretic framework of Abramsky and Brandenburger classifies empirical models by the obstruction to gluing local sections, distinguishing probabilistic, logical, and strong forms of contextuality [26]. The present distinction is cross-cutting: before applying a classicality test, it asks whether one has retained only incidence data, an operational probability fragment, or operator-algebraic product data.

The numerical computations below therefore use two related fragments. First, a deliberately restricted projector-cone proxy uses the listed rank-one projectors as both preparations and effects, with state-side depolarizing noise. Second, for the rational vector sets, a vector-generated operational closure adds unit effects, residual outcomes of projection-valued measures, complements, coarse grainings, and an optional preparation closure by normalized coarse-grained events [17, 18]. The first audit gives exact-certified benchmarks for several projector cones. The second shows how the numbers change when some of the operational data omitted by the proxy are restored.

The detailed examples are deferred to the body of the paper. GHZ and Peres–Mermin are discussed where operator products and masked contexts are analyzed; meager two-valued-state spaces and chromatic colorability are treated in their own combinatorial sections; and the table of thresholds is interpreted only after the computational protocol has been specified. The introduction therefore only fixes the bookkeeping: which data are being kept, and which classicality test is then being applied.

The paper proceeds as follows. Section II defines the three retained structures. Section III recalls the linear-programming test for simplex embeddability, and Section IV gives its simplicial-sandwich interpretation. Section V explains how operator-valued simplex tests depend on whether operators are treated as spectral-outcome data or as algebraic quantities subject to product rules. Section VI explains why two contexts, or weakly pasted contexts, need not by themselves separate vector probabilities from classical convex probabilities. The remaining sections discuss calibration examples, meager two-valued-state spaces, chromatic col-

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orability, the computational protocol, the projector-cone thresholds, and the vector-generated closure comparison.

II. THREE LAYERS OF CONTEXTUALITY

A. Combinatorial contextuality

Combinatorial contextuality is a property of the pasted context structure. Its input is an orthogonality hypergraph $H = (V, \mathcal{C})$ whose vertices are atoms and whose hyperedges are contexts. The basic question is whether there exist global context-independent assignments compatible with the local Boolean structure. The most familiar assignments are two-valued states

$$s : V \to \{0, 1\}, \qquad \sum_{v \in C} s(v) = 1 \quad (C \in \mathcal{C}). \tag{1}$$

Their absence gives the Kochen–Specker (KS) obstruction. Their scarcity gives the meager cases considered below: nonunital, nonseparating, and nonfull two-valued-state spaces. The term *meager* is used here only as a convenient umbrella term for these finite-state-space pathologies; it is not meant in the Baire-category sense. Their abundance, especially a separating family, permits a concrete set representation and hence a partition-logic model [22].

Chromatic contextuality [23] also belongs to the combinatorial level, but it asks for more than the existence of valuations. A strong d-coloring of a d-uniform hypergraph decomposes the constant-one assignment into d two-valued states, one per color. Thus chromatic contextuality is a spectral-labeling obstruction: there is no way to use one fixed set of d nondegenerate outcome labels globally across all contexts. This is a discrete global obstruction, not a probability-table obstruction.

Partition logics—families of partitions, equivalently families of equivalence relations, on a finite set with generalized urn or automaton realizations [22, 27]—sit at the opposite end of the same combinatorial level. Their existence demonstrates that non-Boolean pasting and complementarity can also be represented on an underlying classical point space. Each partition is the set of equivalence classes of an equivalence relation on the underlying set, and a partition logic is obtained by taking a family of such partitions and identifying equal blocks across different partitions. Its dispersion-free states are the point evaluations determined by the underlying elements of that set. In Wright-style generalized urn models, the same construction is realized by ball types and color filters; in Moore or Mealy automata, it is realized by initial states and input-dependent output partitions. Such models implement complementarity without value indefiniteness [27, 28].

B. Simplex contextuality

Simplex contextuality is an operational convex-geometric notion. Its input is not just a hypergraph but a finite prepare-and-measure fragment

$$(\Omega, \mathcal{E}, p(e|\rho)), \tag{2}$$

where Ω is a set of preparations, $\mathcal{E}$ is a set of effects, and $p(e|\rho)$ is the observed probability table. In the quantum cases below, $p(e|\rho) = \mathrm{tr}(e\rho)$ and the preparations and effects are rank-one projectors associated with the listed rays.

A fragment is simplex embeddable if its statistics factor through a classical simplex. Equivalently, the fragment has a generalized-noncontextual explanation in the generalized probabilistic theory (GPT) framework. If no such embedding exists, or if it exists only after adding noise, the fragment is simplex contextual. This property depends on which preparations and effects are included. Enlarging a ray set can change the generated cones and therefore the simplex robustness, even if the additional projectors are merely an orthogonal completion of a previously studied hypergraph.

The two levels can coincide, but need not. A KS set such as the Cabello 18–9 configuration is nonclassical at the valuation level and, for the natural projector fragment, simplex contextual as well. But a partition logic may be fully classical combinatorially while having a quantum double with different Born probabilities.

Conversely, a restricted operational fragment may fail to reveal a defect already present in the underlying hypergraph. For example, nonunitality is exposed only if the offending atom can actually be prepared or measured; nonseparability is exposed only if the fragment contains preparations that statistically distinguish the identified atoms; and a true-implies-false set of two-valued states (TIFS) or nonfullness constraint [29] is exposed only if the relevant endpoint preparations and effects are included. Without such operational witnesses, the simplex test may remain feasible even though the incidence structure has a combinatorial defect.

C. Operator-functional contextuality

Operator-functional contextuality, as the term is used here, is not a new rule beyond Kochen–Specker functional composition. It is a separation of that familiar requirement into two parts. The observables are treated as algebraic quantities, not merely as names for their spectral projectors.

The first is *functional-calculus preservation*:

$$v(f(A)) = f(v(A)). \tag{3}$$

For a sharp observable with spectral resolution

$$A = \sum_i a_i P_i, \tag{4}$$

this condition is largely equivalent to choosing exactly one value-one spectral projector in the resolved Boolean algebra generated by the P_i. When the relevant spectral projectors and their Boolean relations are already part of the combinatorial input, Eq. (3) is therefore mostly reflected in the two-valued-state framework.

The second requirement is *multiplicative preservation* for commuting observables:

$$v(AB) = v(A)v(B), \qquad [A, B] = 0. \tag{5}$$

This is the genuinely operator-algebraic ingredient in GHZ and Mermin parity arguments. It relates different operator presentations that share factors, and it is not captured by looking only at the joint spectral projectors of a single resolved context. Thus the third layer is best understood as having two subcomponents: spectral-functional consistency, which often collapses to projector-valued logic after maximal refinement, and multiplicative consistency, which can impose additional parity constraints across commuting operator products.

This distinction is decisive for GHZ-type arguments. The Mermin–GHZ operators

$$X_1X_2X_3, \quad X_1Y_2Y_3, \quad Y_1X_2Y_3, \quad Y_1Y_2X_3 \tag{6}$$

commute and therefore possess a common eigenbasis. That eigensystem is one orthonormal basis in $\mathbb{C}^8$; as a projector logic it is a single Boolean block with eight atoms and therefore has eight dispersion-free states [21]. There is no KS contradiction at the level of that isolated context. The contradiction enters only when the value of each global product observable is assumed to factor into values of the local Pauli components X_i and Y_i. Then each local factor appears twice in the product of the four assigned values, forcing $+1$ classically, whereas the corresponding operator product is $-\mathbb{1}$. The failure is therefore not the absence of a valuation on one context, but the impossibility of a multiplicative, product-preserving valuation on the operator algebra being implicitly invoked.

The Peres–Mermin square [20, 30, 31] is different but related. At the operator level it is a parity proof. When the masked contexts are extracted by simultaneous diagonalization or by matrix pencils, the operator argument can be transcribed into a projective KS structure, in particular its associated 24–24 (vector–context) completion [21, 32] containing the Cabello–Estebaranz–García-Alcaine 18–9 set [7, 21]. Thus some multiplicative operator contradictions descend to combinatorial projector contextuality after unmasking, while others, such as the single-context GHZ eigensystem, do not. This is why the operator-functional layer deserves to be kept separate from both colorability and simplex embeddability, while still recognizing that its functional-calculus part overlaps with the projector-valued combinatorial layer after full spectral resolution.

The word "single-context" is therefore a statement about one particular operational packaging, not about the full GHZ reasoning. The collective measurement of the four global products is a single Boolean context. The local GHZ experiment, however, invokes four incompatible local settings and the counterfactual identification of local factors across them. In the first packaging the simplex and two-valued-state tests are trivial; in the second packaging the nonclassicality is absolute and rests on locality, noncontextuality, and product preservation. Equivalently, the collective product observable and its local-factor implementation should not be identified without further assumptions. The former reports a global spectral outcome; the latter refines that outcome into local Pauli outcomes and then imposes a product rule on their assigned values.

D. Probability theories on the same pasting

The probability level is itself not exhausted by the binary alternative "classical or quantum." On a fixed exclusivity structure one may consider at least three types of context-independent additive weights. First, classical probabilities are convex mixtures of dispersion-free states; they form a polytope. Second, Born probabilities arise from a faithful orthogonal representation and a state vector or density operator; graph-theoretically, these are related to Lovász-type theta-body constructions. Third, there are exotic weights, such as Wright-type weights on pentagon-like logics [33, 34], which satisfy additivity within contexts but are neither classical convex-hull points nor Hilbert-space Born probabilities.

Thus the incidence structure alone does not determine the probability calculus. The same pasted logic may admit a partition model, a vector representation, and additional exotic additive weights. This is the sense in which combinatorial contextuality and simplex contextuality are largely independent diagnostics. The former classifies global assignments on the logic; the latter tests a particular operational probability model.

III. THE OPERATIONAL SIMPLEX TEST

Let $\Omega = \{\rho_1, \dots, \rho_N\}$ be a finite set of quantum states, and let $\mathcal{E} = \{e_1, \dots, e_M\}$ be a finite set of quantum effects. The states may be normalized or subnormalized. Here subnormalized means

$$\rho_i \geq 0, \qquad 0 \leq \mathrm{tr}(\rho_i) \leq 1. \tag{7}$$

Thus one may write $\rho_i = p_i\widehat{\rho}_i$, where $\widehat{\rho}_i$ is a normalized density operator and $0 \leq p_i \leq 1$ is a weight or success probability. This convention is natural for cone constructions, since the cones below are closed under nonnegative rescaling.

Let $\mathcal{H}$ be the finite-dimensional Hilbert space of the fragment, and let

$$\mathcal{A} = \mathrm{Herm}(\mathcal{H}) \tag{8}$$

denote the real vector space of Hermitian operators on $\mathcal{H}$. We do not assume that the chosen coordinates on $\mathcal{A}$ are trace-orthonormal. Instead, choose an arbitrary real basis

$$B_1, \dots, B_q \tag{9}$$

of $\mathcal{A}$. If

$$A = \sum_{\alpha=1}^{q} a_\alpha B_\alpha, \qquad C = \sum_{\beta=1}^{q} c_\beta B_\beta, \tag{10}$$

then the trace pairing is represented by the Gram matrix

$$G_{\alpha\beta} = \mathrm{tr}(B_\alpha B_\beta). \tag{11}$$

Thus, in these coordinates,

$$\mathrm{tr}(AC) = a^T G c. \tag{12}$$

Only in a trace-orthonormal basis would G be the identity matrix.

Let

$$S_\Omega = \mathrm{span}(\Omega), \qquad S_E = \mathrm{span}(\mathcal{E}) \tag{13}$$

be the accessible state and effect subspaces. Choose arbitrary real bases of S_Ω and S_E. Let

$$d_\Omega = \dim S_\Omega, \qquad d_E = \dim S_E. \tag{14}$$

The inclusion maps

$$I_\Omega : S_\Omega \hookrightarrow \mathcal{A}, \qquad I_E : S_E \hookrightarrow \mathcal{A} \tag{15}$$

are the coordinate embeddings determined by these choices of bases. Concretely, I_Ω is a $q \times d_\Omega$ matrix whose columns are the ambient coordinates, in the basis $B_1, \dots, B_q$, of the chosen basis vectors of S_Ω. Similarly, I_E is a $q \times d_E$ matrix whose columns are the ambient coordinates of the chosen basis vectors of S_E.

Therefore, if $x \in \mathbb{R}^{d_\Omega}$ is the coordinate vector of an accessible state and $y \in \mathbb{R}^{d_E}$ is the coordinate vector of an accessible effect, then their ambient coordinate vectors are $I_\Omega x$ and $I_E y$, respectively. The Born pairing between them is

$$\mathrm{tr}(e_y \rho_x) = (I_E y)^T G (I_\Omega x) = y^T I_E^T G I_\Omega x. \tag{16}$$

Thus

$$B_{E\Omega} = I_E^T G I_\Omega \tag{17}$$

is the matrix of the accessible state–effect pairing in the chosen reduced coordinates.

After choosing the reduced bases of S_Ω and S_E, let V_Ω and V_E denote the matrices whose columns are the coordinate vectors of the ρ_i and e_j, respectively. The state and effect cones are first given by their generator, or V-, representations:

$$\begin{aligned}\mathrm{Cone}[\Omega] &= \left\{ V_\Omega r : r \in \mathbb{R}^N,\ r_i \geq 0 \right\} \\ &= \left\{ \sum_{i=1}^{N} r_i \rho_i : r_i \geq 0 \right\} \subseteq S_\Omega, \end{aligned} \tag{18}$$

$$\begin{aligned}\mathrm{Cone}[\mathcal{E}] &= \left\{ V_E s : s \in \mathbb{R}^M,\ s_j \geq 0 \right\} \\ &= \left\{ \sum_{j=1}^{M} s_j e_j : s_j \geq 0 \right\} \subseteq S_E. \end{aligned} \tag{19}$$

Because these cones are finitely generated, they are polyhedral. Hence they also admit half-space, or H-, representations, as by the Minkowski-Weyl theorem [35–41] every polytope has equivalent descriptions, by finite sets of vertices (V-representation) and of facet (in)equalities (H-representation). This stage is similar to the standard computation of Bell-type inequalities, like the Clauser–Horne–Shimony–Holt inequalities [10, 12]. Those H-representations are linear functionals

$$h_1^\Omega, \dots, h_{n_\Omega}^\Omega \in S_\Omega^*, \qquad h_1^E, \dots, h_{n_E}^E \in S_E^*, \tag{20}$$

which may be chosen as facet-defining inequalities, such that

$$\mathrm{Cone}[\Omega] = \left\{ x \in S_\Omega : h_k^\Omega(x) \geq 0 \text{ for } k = 1, \dots, n_\Omega \right\}, \tag{21}$$

$$\mathrm{Cone}[\mathcal{E}] = \left\{ y \in S_E : h_\ell^E(y) \geq 0 \text{ for } \ell = 1, \dots, n_E \right\}. \tag{22}$$

Equivalently, collect these functionals as the rows of linear maps

$$H_\Omega : S_\Omega \to \mathbb{R}^{n_\Omega}, \qquad H_E : S_E \to \mathbb{R}^{n_E}, \tag{23}$$

so that

$$H_\Omega x = \begin{pmatrix} h_1^\Omega(x) \\ \vdots \\ h_{n_\Omega}^\Omega(x) \end{pmatrix}, \qquad H_E y = \begin{pmatrix} h_1^E(y) \\ \vdots \\ h_{n_E}^E(y) \end{pmatrix}. \tag{24}$$

With this notation,

$$\begin{aligned} H_\Omega x \geq_{\mathrm{e}} 0 &\iff x \in \mathrm{Cone}[\Omega], \\ H_E y \geq_{\mathrm{e}} 0 &\iff y \in \mathrm{Cone}[\mathcal{E}]. \end{aligned} \tag{25}$$

Here $\geq_{\mathrm{e}}$ denotes entrywise nonnegativity: all components of the corresponding vector must be nonnegative. It is not the positive-semidefinite order on Hermitian operators.

The simplex-embeddability linear program asks whether the Born-pairing matrix (17) factors through the facet descriptions of the state and effect cones. Equivalently, one asks whether there exists an entrywise nonnegative matrix

$$\sigma \in \mathbb{R}^{n_E \times n_\Omega} \tag{26}$$

such that

$$I_E^T G I_\Omega = H_E^T \sigma H_\Omega, \qquad \sigma \geq_{\rm e} 0. \tag{27}$$

The dimensions are

$$I_E^T G I_\Omega \in \mathbb{R}^{d_E \times d_\Omega}, \qquad H_E^T \sigma H_\Omega \in \mathbb{R}^{d_E \times d_\Omega}. \tag{28}$$

Thus both sides represent the same bilinear pairing between accessible effects and accessible states.

When Eq. (27) is feasible, the Born statistics generated by all pairs $(\rho, e) \in \Omega \times \mathcal{E}$ admit a simplex-embeddable classical explanation for the specified fragment. This is the sense in which the fragment is classical and "noncontextual" in the simplex test. It should not be confused with the stronger KS requirement of a global two-valued valuation on an independently specified hypergraph.

The normalization constraints are contained in the operational fragment rather than in an additional free normalization equation for σ. The unit effect, when included, is carried through the same accessible-coordinate maps I_Ω and I_E; the reconstructed simplex states are normalized by requiring the embedded unit effect to evaluate to one on them, and response functions are bounded by the included effect cone and its coarse-grainings. Consequently Eq. (27) is the cone-factorization form of the simplex-embeddability criterion for the specified accessible GPT fragment. It becomes the full operational Schmid–Selby–Wolfe–Kunjwal–Spekkens test only when the full operational scenario—unit effects, complements, coarse grainings, preparation and measurement-event equivalences, and normalization data—has been supplied.

If it is infeasible, one may introduce an operational noise map

$$D : \mathcal{A} \to \mathcal{A} \tag{29}$$

on the state side and minimize r subject to

$$\begin{aligned} I_E^T G \big[(1-r)\,\mathrm{Id}_{\mathcal{A}} + rD\big] I_\Omega &= H_E^T \sigma H_\Omega, \\ \sigma \geq_{\rm e} 0, \qquad 0 &\leq r \leq 1. \end{aligned} \tag{30}$$

Here $\mathrm{Id}_{\mathcal{A}}$ denotes the identity map on the ambient Hermitian-operator coordinate space. The optimum r is a robustness against the specified noise. In the calculations below, D is always the completely depolarizing state-side map

$$\rho \longmapsto \mathrm{tr}(\rho) \frac{\mathbb{1}_{d_H}}{d_H}, \tag{31}$$

where $d_H = \dim \mathcal{H}$. Although Eq. (31) is written for positive operators, in Eq. (30) it is used as its unique linear extension to the ambient real vector space $\mathcal{A}$ of Hermitian operators.

Two qualifications fix the scope of the restricted tests used below. First, the linear program is not a hypergraph-coloring or two-valued-state test. In the projector-cone version, the context structure is seen only through the operator geometry of the listed projectors: orthogonality, linear dependence, and relations such as $\sum_{v \in C} P_v = \mathbb{1}$ when a complete context is present. The program does not impose the deterministic rule that exactly one atom in each block is true.

Second, simplex embeddability is weaker than simpliciality. A generalized probabilistic theory or an accessible fragment may embed into a classical simplex even though its own state space is not itself a simplex. This distinction is central to the generalized-probabilistic-theory analysis of Schmid *et al.* [17]. For accessible fragments, cone equivalence is especially important: cone-equivalent fragments are either both classically explainable or both not, because the embedding problem depends on the generated cones [42].

Finally, Eq. (27) is not the same object as the convex hull of two-valued states of an orthogonality hypergraph. The two-valued-state hull appears as a sharp, deterministic shadow of the operational problem when the fragment contains the eigenstate preparations, sharp projective measurements, and operational equivalences that force deterministic responses on the relevant ontic support. This is the bridge to KS reasoning, not an identity of the two frameworks.

The computations reported in Table I then impose a deliberately restricted *projector-cone* version of the simplex test. The listed rank-one projectors are used as preparations and as effects, and the cone generated by those projectors is tested against state-side depolarizing noise; unit effects, complementary coarse-grainings, and context-induced operational equivalences are not added unless they are generated by the same projector cone. Thus two presentations with the same projector set but different groupings into contexts define the same problem. The thresholds below are consequently benchmarks for these projector cones, not for complete operational scenarios.

IV. THE SIMPLICIAL-SANDWICH CRITERION AND USEFUL DIAGNOSTICS

The exact simplex criterion can be stated compactly as follows. A finite prepare-and-measure fragment is classical at the simplex level exactly when its accessible Born pairing factors through a finite classical orthant. In the coordinate convention of Section III, this Born pairing is the bilinear form

$$(y, x) \longmapsto y^T I_E^T G I_\Omega x. \tag{32}$$

Thus, after quotienting by operational equivalences, there must exist a finite ontic set Λ and positive linear maps

$$\mu : S_\Omega \longrightarrow \mathbb{R}_+^\Lambda, \qquad \xi : S_E \longrightarrow \mathbb{R}_+^\Lambda, \tag{33}$$

such that, for all included preparations and effects,

$$\mathrm{tr}(e\rho) = \sum_{\lambda \in \Lambda} \xi_e(\lambda) \mu_\rho(\lambda), \tag{34}$$

with the usual normalization and response constraints. Thus the classical simplex appears as the normalized slice of the orthant $\mathbb{R}_+^\Lambda$. In this language, simplex nonclassicality is precisely the failure of the positive factorization in Eq. (34) for the specified fragment.

Equivalently, once the effect cone is identified with functionals on the accessible state space through the Born pairing $I_E^T G I_\Omega$, the problem may be pictured as a sandwiching problem. One asks whether there is a simplicial cone K such that

$$\mathrm{Cone}(\Omega_A) \subseteq K \subseteq \mathrm{Cone}(E_A)^*, \tag{35}$$

where the dual cone is taken with respect to the same Born pairing, not with respect to an arbitrary Euclidean dot product. The cone K is the unnormalized classical state cone of a hidden-variable model, while K^* supplies the positive response functionals. In the facet representation this sandwich condition is exactly the nonnegative matrix equation (27); with noise it becomes Eq. (30). By Farkas duality, failure of the sandwich is witnessed by a separating linear functional, equivalently by a noncontextuality inequality for the chosen operational table.

This geometric picture also explains why the simplex test is not a graph invariant. The orthogonality hypergraph records exclusivity and context incidence, but the simplex linear program sees the accessible cones, the Born bilinear form, and the operational equivalences. Metric information such as $|\langle\psi|\phi\rangle|^2$ matters. Conversely, as has been noted before, context grouping is invisible to a projector-only cone test unless it is encoded by additional effects, coarse grainings, or equivalence constraints. The exact question is therefore always: can this specified operational fragment, with these equivalences, be sandwiched by a simplicial cone?

The following diagnostics are useful, but they should not be mistaken for necessary-and-sufficient graph criteria. First, a fragment whose accessible state cone is already simplicial is normally classically harmless; nonsimpliciality or overcompleteness of the generated cone is a warning sign. But merely having many extreme rays is not by itself an obstruction, since a nonsimplicial cone may still lie inside a larger simplicial cone compatible with the effects. Second, the effects must resolve the nonsimplicial geometry. Coarse or incomplete effects can make a real obstruction invisible. Third, the constraints must close a nontrivial consistency loop: cyclic exclusivity, TIFS or KS implication chains, parity structures after spectral refinement, or equal-mixture preparation equivalences. Without such a loop the hidden-variable sample space often has enough freedom to absorb the transition probabilities.

Three common mechanisms are worth separating. An *overlap or cyclic-exclusivity obstruction* appears when nonorthogonal quantum states and exclusive effects must be reconciled around a closed cycle; the completed pentagon is the canonical low-dimensional calibration. A *determinism obstruction* appears when eigenstate preparations and sharp effects force response functions to take values 0 or 1 on large parts of the ontic space; in sufficiently interlocked configurations this is the operational shadow of KS, TIFS, and related two-valued-state constraints. A *convex-decomposition obstruction* appears when different accessible mixtures represent the same operational preparation and hence must be assigned the same ontic distribution. This last mechanism is the usual preparation-contextuality mechanism and can occur in small scenarios if the relevant mixture equivalences are included.

These mechanisms illuminate the examples below. The firefly logic L_{12} [43] realizes complementarity but does not produce a simplex obstruction in the restricted projector-cone fragment [27, 28]. The completed Klyachko–Can–Binicioğlu–Shumovsky pentagon already exhibits a cyclic probability-level gap, although it is not KS uncolorable [13, 34]. Meager-state configurations expose deterministic two-valued-state defects in different ways [44]. The original Yu–Oh 13-ray set [14] and its 25-ray orthogonal completion used here, the Cabello–Estebaranz–García-Alcaine 18–9 set [7], and the Peres–Mermin square together with the 24–24 completion [20, 21, 30–32] give increasingly constrained projector-cone fragments whose Born pairings fail simplex embeddability without noise. GHZ is different: if one retains only the common joint spectral projectors of the four commuting global product observables, the fragment is a single Boolean context and hence simplex-classical. Its nonclassicality enters through the additional requirement that values of global products factor into pre-existing local Pauli values, a product-preservation constraint not present in the projector-only sandwich [19–21].

V. OPERATOR-VALUED SIMPLEX TESTS AND MASKED CONTEXTS

An “operator-valued simplex” is not a single unambiguous object. Its meaning depends on what is taken as the operational data and what constraints are imposed on the deterministic classical vertices.

First, if an operator is used only as shorthand for its spectral projectors, then there is no essential change from the projector simplex. A normal observable

$$A = \sum_i a_i P_i \tag{36}$$

contributes expectation values

$$\langle A\rangle_\rho = \sum_i a_i \ \mathrm{tr}(\rho P_i), \tag{37}$$

which are linear post-processings of the spectral-outcome probabilities. If the projectors P_i are already included as effects, adding A itself changes only the coordinates of the data table.

Second, if one keeps only a degenerate operator and does not include enough commuting observables to resolve its eigenspaces, then the simplex test becomes coarser. Some distinctions between projectors are hidden. This can make simplex embeddability easier and may miss contextuality that appears after maximal refinement. The matrix-pencil method addresses precisely this issue: a suitable linear combination of mutually commuting degenerate operators reveals the common spectral projectors and hence the underlying context [21].

Third, if the classical model is required to assign values to operators while preserving functional and product relations, the test changes substantially. The deterministic vertices are no longer arbitrary assignments to spectral outcomes; they must also satisfy relations such as Eq. (5). This is the setting of Peres–Mermin and GHZ parity arguments. In such cases the obstruction may be algebraic even when a projector-only simplex for a single context would be trivial.

Consequently, an operator-valued simplex can mean at least three different things: a simplex for expectation values, a simplex for spectral-outcome probabilities, or a simplex whose vertices are deterministic operator value assignments constrained by functional calculus and, where applicable, by multiplicative product rules. The first two are operational probability tests. The last is an operator-functional test, and the GHZ ingredient is specifically the product-rule part, not merely functional calculus on a single spectral resolution. This is the sense in which GHZ, although a single context after diagonalization, still violates classical expectations: the violation is not in the Boolean algebra of its joint spectral projectors but in the attempted decomposition of its global product observables into context-independent local elements of reality.

A. Consecutive global measurements versus local factorizations

One might object that the four GHZ product observables commute. Why not simply measure them consecutively on the same three-particle system? In notation, the four observables are

$$\begin{aligned} A_1 &= X_1X_2X_3, & A_2 &= X_1Y_2Y_3, \\ A_3 &= Y_1X_2Y_3, & A_4 &= Y_1Y_2X_3. \end{aligned} \tag{38}$$

This consecutive protocol is possible in principle. Since the A_i commute, their spectral projections commute. An ideal Lüders measurement of one product observable collapses the state only to an eigenspace that is invariant under all the others. Subsequent ideal measurements refine the state to a common eigenspace and do not erase the previously registered eigenvalues. Equivalently, one may perform the joint projective measurement with atoms

$$P_{\boldsymbol{\varepsilon}} = \prod_{i=1}^{4} \frac{\mathbb{1} + \varepsilon_i A_i}{2}, \qquad \varepsilon_i \in \{\pm 1\}, \qquad \varepsilon_1\varepsilon_2\varepsilon_3\varepsilon_4 = -1, \tag{39}$$

where only eight sign patterns are nonzero because $A_1A_2A_3A_4 = -\mathbb{1}$. Thus a collective consecutive protocol registers four repeatable outcomes whose product is always -1.

Remark 1 (Commuting observables and registered outcomes)**.** *The preceding statement uses the standard finite-dimensional spectral fact that commuting self-adjoint observables admit compatible Lüders measurements. Let A and B be self-adjoint operators with $[A, B] = 0$, and let P_a and Q_b be their spectral projectors. Since spectral projectors are obtained from the operators by functional calculus, $[P_a, Q_b] = 0$ for all eigenvalues a, b. Hence $BP_a = P_aB$, so the subspace $P_a\mathcal{H}$ is invariant under B. Equivalently,*

$$P_a\mathcal{H} = \bigoplus_b P_aQ_b\mathcal{H}, \tag{40}$$

up to zero summands. Thus the collapse caused by registering the outcome a of A does not cut across the eigenspace decomposition of B; it only selects a direct sum of joint eigenspaces.

For a Lüders measurement of A followed by a measurement of B, the sequential joint probability is

$$p(a \text{ then } b) = \operatorname{tr}\big(Q_bP_a\rho P_a\big) = \operatorname{tr}\big(P_aQ_b\rho\big),$$

which is precisely the probability assigned by the simultaneous joint projection-valued measure (PVM) $\{P_aQ_b\}_{a,b}$. A later measurement of B may refine the post-measurement state within $P_a\mathcal{H}$, but it cannot invalidate the previously registered value a. This proof applies to the degenerate observable itself. It does not apply to an arbitrary finer apparatus that, while reporting the same coarse outcome a, measures additional degrees of freedom inside $P_a\mathcal{H}$ which need not commute with B.

This collective protocol does not, by itself, yield a GHZ contradiction. It is just one joint measurement, hence one Boolean context. A classical hidden-variable model for it is immediate: the ontic state is one of the eight allowed sign patterns $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4)$ satisfying $\prod_i \varepsilon_i = -1$, and the measurement reads off its four coordinates. No product rule for local factors is used, and no inconsistency follows.

The GHZ contradiction appears only if the same global product values are required to arise from pre-existing local Pauli values, $v(X_i), v(Y_i) \in \{\pm 1\}$, $v(X_1Y_2Y_3) = v(X_1)v(Y_2)v(Y_3)$, with analogous equations for the other three products. Multiplying the four classical equations gives $+1$, because every local factor occurs twice. Quantum mechanically the product of the four commuting global operators is $-\mathbb{1}$. The collective measurement of Eq. (39) reveals the global products, but it never reveals the six local quantities X_i, Y_i whose context-independent coexistence is being assumed. In that precise sense the collective protocol is factor-blind.

The local GHZ protocol operationalizes a different refinement. In each run the three separated parties choose one of the four compatible triples $\{X_1, X_2, X_3\}$, $\{X_1, Y_2, Y_3\}$, $\{Y_1, X_2, Y_3\}$, $\{Y_1, Y_2, X_3\}$, and multiply the local outcomes. These four triples are not jointly measurable by local sharp measurements, since on each particle X_i and Y_i anticommute. Locality, or equivalently the Einstein–Podolsky–Rosen (EPR) counterfactual step in the usual GHZ reasoning, is what motivates assigning the same value to X_i or Y_i independently of which compatible triple is chosen. The contradiction therefore concerns the attempted identification of two operationally different realizations of a product observable: a collective entangled measurement of the product as a whole, and a local measurement of its factors followed by multiplication.

There is a final technical caveat. A degenerate observable does not determine a unique measurement apparatus. A badly chosen implementation may disturb states inside a degenerate eigenspace by effectively measuring additional, possibly incompatible, degrees of freedom. That is not a failure of commutativity of the abstract observables; it is a different physical refinement of the degenerate measurement. For the ideal consecutive protocol considered here, one assumes Lüders instruments or the common joint PVM, in which case the registered product outcomes are compatible and repeatable. What remains nonclassical is not the sequence of global product outcomes, but the extra product-preserving identification of those outcomes with jointly pre-existing local values.

B. Single-context triviality, unmasking, and the chromatic analogy

The preceding discussion also fixes a possible misconception about the scope of simplex tests. A closed fragment consisting of one joint spectral measurement is always simplex embeddable. Its ontic states may simply be the atoms of that Boolean algebra, and an arbitrary preparation is represented by the corresponding probability distribution over those atoms. Thus the collective GHZ product fragment, taken by itself, has no projector-simplex obstruction. This is not a positive classical explanation of the full GHZ experiment; it is only the trivial simplex embeddability of a closed single Boolean context.

The qualifier is essential. One should not conclude that operator-valued arguments can never enter a simplex analysis. They can do so whenever the operator proof is unfolded into a multi-context projector fragment, or whenever the operational scenario includes the local factors and their compatibility relations. Peres–Mermin supplies the clearest example: its algebraic parity proof, once the degenerate observables are resolved by simultaneous diagonalization or matrix pencils, yields a genuine multi-context projective configuration, and that configuration can be subjected to two-valued-state and simplex tests. The GHZ collective context does not itself have this property; the local GHZ experiment has a different operational structure, namely four incompatible local settings together with the product-rule identification of shared local factors.

The operative distinction is therefore not whether an argument is operator-valued or simplex-theoretic in the abstract. The right question is whether the retained fragment is merely a single collective spectral context or an unmasked multi-context fragment. The former is classically representable as a Boolean block. The latter may display ordinary projector contextuality, simplex nonembeddability, Bell–GHZ nonlocality, or an operator-functional parity contradiction, depending on which structures are retained in the fragment.

This also clarifies the relation to chromatic contextuality. GHZ and strong chromatic contextuality are similar only in the negative sense that neither is exhausted by a probability-table simplex test. Positively they are different obstructions. Chromatic contextuality is combinatorial: it asks whether a fixed nondegenerate spectrum, or equivalently a strong coloring, can be assigned globally across a hypergraph. GHZ is algebraic: it asks whether values can preserve products of commuting observables and their local factors. State-independent parity proofs such as Peres–Mermin are closer in spirit to chromatic obstructions because they express a global spectral inconsistency independent of the input state, but the mechanism is still operator multiplication rather than hypergraph coloring. Thus both belong outside the simplex-probability axis, but they lie on different nonsimplex axes: one incidence/coloring-theoretic, the other operator-algebraic.

VI. WHY TWO CONTEXTS NEED NOT, BY THEMSELVES, REVEAL NONCLASSICALITY

Before turning to calibration examples, it is useful to remove one smaller temptation: the idea that the passage from one context to two already separates quantum vector probabilities from classical convex probabilities. That intuition is correct at the level of the full theories but misleading for a small operational fragment. A single orthonormal context is an ordinary probability simplex. Two nonintertwining contexts are, operationally, a pair of simplexes connected by a stochastic transition matrix.

Let

$$A = \{a_1, \dots, a_d\}, \qquad B = \{b_1, \dots, b_d\} \tag{41}$$

be two orthonormal bases. The quantum transition matrix

$$T_{ji} = |\langle b_j | a_i \rangle|^2 \tag{42}$$

is doubly stochastic. For the restricted fragment consisting only of preparations in $A \cup B$ and measurements in $A \cup B$, this matrix can be reproduced by a classical random channel. One may take ontic states to be pairs $\lambda = (i, j)$, where i is the predetermined answer to context A and j is the predetermined answer to context B.

Preparing a_i means loading the classical ensemble with the states $(i,1),\dots,(i,d)$ in proportions $T_{1i},\dots,T_{di}$. A measurement of A reads the first coordinate and therefore returns i with certainty; a measurement of B reads the second coordinate and returns the Born transition probabilities. Preparations b_j are treated symmetrically. No independence assumption is being imposed on two physical subsystems here; $\lambda=(i,j)$ is only a bookkeeping device for predetermined answers in two otherwise disconnected measurement contexts. The construction works precisely because no further cross-context consistency constraints are present.

This construction is not a classical simulation of the full Hilbert-space theory. It does not reproduce arbitrary third bases, interference phases, sequential state update, or the entire lattice of subspaces. Nor is it a theorem that every possible two-context operational scenario is simplex embeddable: if additional preparations, operational equivalences, or linear dependences among preparations and effects are imposed, preparation contextuality may already be witnessed in very small scenarios, including scenarios with two binary measurements [45]. The limited claim here is that a bare transition table between two bases, with no further operational equivalences beyond normalization, can be represented as an ordinary stochastic channel. Nonclassicality becomes visible when one asks for a single classical sample space to satisfy enough compatibility constraints around a nontrivial web of pasted contexts, or when additional equivalences constrain the preparations and effects. The obstruction is not the mere fact of having two contexts, but the consistency requirements imposed on them.

This is precisely the lesson of partition logics. A partition logic is obtained by pasting Boolean algebras arising from partitions of a finite set. Its atoms can be represented as subsets of a common classical sample space, and probabilities are ordinary convex mixtures over dispersion-free states. Equivalently, the same structures can be modeled by Wright-style generalized urns or by the initial-state identification problem for finite deterministic Moore or Mealy automata. In a generalized urn, a ball type is an ontic state and different colors correspond to incompatible experimental contexts; looking through one color filter reveals one partition while hiding the others. In the automaton model, the initial state is the ontic state and different input symbols induce different observable partitions of the state set. These models realize complementarity without value indefiniteness.

The contrast with a Hilbert-space realization is then not in the exclusivity graph alone. The same graph may support a set-theoretic partition representation, a faithful orthogonal vector representation, or even more exotic weights [27]. The probability type is determined by the chosen physical representation and the operational fragment, not by the bare pasting diagram alone. Thus the statement "complementarity does not imply contextuality" is not merely philosophical: it is a concrete modeling fact.

VII. CALIBRATION EXAMPLES: FIREFLY LOGIC AND THE PENTAGON

The first calibration example is the smallest nontrivial pasting, the firefly logic L_{12} is a 5–2 vertex–context set, obtained by pasting two three-atomic Boolean algebras along one common atom [43],

$$\{a,b,c\}, \qquad \{a,d,e\}. \tag{43}$$

A faithful qutrit realization is

$$a=(1,0,0), \quad b=(0,1,0), \quad c=(0,0,1), \tag{44}$$

$$d=(0,1,1), \quad e=(0,1,-1). \tag{45}$$

The partition-logic representation is even more elementary. On the finite set $S_5=\{1,\dots,5\}$ one may use the two partitions

$$\big\{\{2,3\},\{4,5\},\{1\}\big\}, \qquad \big\{\{1\},\{3,5\},\{2,4\}\big\}, \tag{46}$$

which paste along the singleton atom $\{1\}$. This is a generalized urn or finite-automaton realization of the same logic. It is non-Boolean as a pasted structure, but it is still classically representable. Its role here is diagnostic: it is the first place where contexts are not simply disjoint, yet simplex nonembeddability should not be expected merely from that fact.

The next calibration example is the pentagon, or pentagram logic. In the Klyachko–Can–Binicioğlu–Shumovsky (KCBS) form one starts with five qutrit rays v_i satisfying cyclic orthogonality

$$v_i \perp v_{i+1} \qquad (i \bmod 5). \tag{47}$$

A convenient realization is

$$v_k=\left(\cos\frac{4\pi k}{5},\, \sin\frac{4\pi k}{5},\, \sqrt{\cos\frac{\pi}{5}}\right), \qquad k=0,\dots,4. \tag{48}$$

Completing each orthogonal pair by $w_i=v_i\times v_{i+1}$ gives five complete qutrit contexts

$$\{v_i,w_i,v_{i+1}\}, \qquad i=0,\dots,4, \tag{49}$$

and hence a 10-ray pentagon (10–5 vertex–context set).

The pentagon sits between L_{12} and the Specker bug discussed later. It is still partition-logically representable: the five cyclically intertwined contexts support 11 two-valued states, from which a partition logic over $S_{11}=\{1,\dots,11\}$ can be reconstructed; explicitly, one may take the 11 admissible two-valued states as the underlying points and represent each atom by the subset of states assigning value one to it. Nevertheless, the pentagon already displays a difference between classical convex probabilities and Born/Lovász-type vector probabilities. On the five outer atoms, the classical convex hull yields the familiar pentagon bound $\sum_i p_i \le 2$, whereas the optimal faithful orthogonal vector representation gives the quantum value $\sqrt{5}$. This is the first useful calibration point at which cyclic exclusivity exposes a quantitative difference, though still not a KS impossibility.

VIII. COMBINATORIAL CONTEXTUALITY: TWO-VALUED STATES AND MEAGERNESS

Let $H = (V, \mathcal{C})$ be a finite d-uniform orthogonality hypergraph. The vertices $v \in V$ represent atoms, usually rank-one projectors, and each context $C \in \mathcal{C}$ is a d-element set of mutually orthogonal atoms corresponding to a maximal sharp measurement. A faithful orthogonal representation in $\mathbb{C}^d$ or $\mathbb{R}^d$ assigns to every vertex a ray $|v\rangle$ such that vertices are orthogonal precisely when the hypergraph says they are, and every context resolves the identity:

$$\sum_{v\in C} \Pi_v = \mathbb{1}, \qquad \Pi_v = \frac{|v\rangle\langle v|}{\langle v|v\rangle}. \tag{50}$$

Definition 1 (Two-valued state)**.** *A two-valued state on H is a map $s : V \to \{0, 1\}$ such that*

$$\sum_{v\in C} s(v) = 1 \qquad \text{for every } C \in \mathcal{C}. \tag{51}$$

We denote the set of all such states by $\mathcal{S}_2(H)$.

The deterministic valuation polytope is

$$\mathsf{P}_2(H) = \operatorname{conv} \mathcal{S}_2(H) \subseteq [0, 1]^V. \tag{52}$$

It is useful to distinguish several ways in which $\mathcal{S}_2(H)$ can be deficient. Tkadlec called attention to hypergraphs whose state spaces are empty, not unital, not separating, or not full [44]; these notions are standard in the quantum-logic literature [28, 46, 47].

Definition 2 (Unital, separating, full)**.** *Let $\mathcal{S}_2(H)$ be the set of two-valued states on H. It is called*

1. unital *if for every vertex $a \in V$ there exists $s \in \mathcal{S}_2(H)$ with $s(a) = 1$;*
2. separating *if for every distinct pair $a, b \in V$ there exists $s \in \mathcal{S}_2(H)$ with $s(a) \neq s(b)$;*
3. full *if for every nonorthogonal pair $a, b \in V$ there exists $s \in \mathcal{S}_2(H)$ with $s(a) = s(b) = 1$.*

In the usual orthomodular-poset formulation, where atoms sit inside a poset with zero, unit, and orthocomplementation, fullness implies separation and separation implies unitality. In the atom-only hypergraph formulation used here these implications require the corresponding reconstruction assumptions. We therefore treat unitality, separation, and fullness as separate diagnostics and do not rely on the hierarchy without those assumptions.

Definition 3 (Meager two-valued-state space)**.** *A hypergraph has a meager two-valued-state space if $\mathcal{S}_2(H)$ is empty, nonunital, nonseparating, or nonfull. This is an umbrella term; the four cases have different operational meanings.*

Each failure mode corresponds to a different linear constraint on $\mathsf{P}_2(H)$. If $\mathcal{S}_2(H) = \emptyset$, then $\mathsf{P}_2(H) = \emptyset$. This is the KS case. If $\mathcal{S}_2(H)$ is nonunital at a, then

$$x_a = 0 \qquad \text{for all } x \in \mathsf{P}_2(H). \tag{53}$$

If it is nonseparating for $a \neq b$, meaning $s(a) = s(b)$ for all $s \in \mathcal{S}_2(H)$, then

$$x_a = x_b \qquad \text{for all } x \in \mathsf{P}_2(H). \tag{54}$$

If it is nonfull for a nonorthogonal pair a, b, meaning no s has $s(a) = s(b) = 1$, then

$$x_a + x_b \leq 1 \qquad \text{for all } x \in \mathsf{P}_2(H). \tag{55}$$

A true-implies-false (TIFS) structure, such as the Specker bug, is the stronger directed version: for a nonorthogonal pair a, b,

$$s(a) = 1 \Longrightarrow s(b) = 0, \tag{56}$$

for all admissible s. Equation (56) implies that the face $x_a = 1$ of the two-valued-state polytope is contained in $x_b = 0$.

IX. HOW MEAGERNESS BECOMES OPERATIONAL

The preceding notions are combinatorial: they concern the possible two-valued states on a hypergraph. To become operational, however, such defects must be visible in the actual preparations and effects included in the fragment. A logical obstruction that is present in the hypergraph may remain invisible if the fragment is too poor to test it.

Let H have a faithful orthogonal representation, with vertex projectors Π_v. For a preparation ρ, write

$$p_v^\rho = \operatorname{tr}(\rho \Pi_v) \tag{57}$$

for the Born probability assigned to the vertex effect Π_v. The deterministic two-valued-state polytope

$$\mathsf{P}_2(H) = \operatorname{conv} \mathcal{S}_2(H) \tag{58}$$

contains exactly the probability assignments obtainable as classical mixtures of admissible two-valued states. Thus, if an accessible Born vector p^ρ lies outside $\mathsf{P}_2(H)$, then no deterministic two-valued-state model can reproduce that part of the fragment.

The different forms of meagerness become visible in different ways. If the two-valued states are nonunital at an atom a, then every admissible two-valued state assigns

$$s(a) = 0. \tag{59}$$

Consequently every classical mixture in $\mathsf{P}_2(H)$ also assigns probability zero to a. But if the fragment allows the eigenstate preparation $\rho = \Pi_a$ and the effect Π_a, quantum theory gives

$$\mathrm{tr}(\Pi_a \Pi_a) = 1. \tag{60}$$

The atom that is never true in the two-valued-state model is obtained with certainty in the corresponding quantum eigenstate. Nonunitality is then directly operationally visible.

If the two-valued states are nonseparating for two distinct atoms a and b, then every admissible valuation identifies them:

$$s(a) = s(b). \tag{61}$$

Every classical mixture therefore satisfies $x_a = x_b$. This becomes operationally relevant only if the Born statistics can distinguish the two effects, that is, if the fragment contains some preparation ρ such that

$$\mathrm{tr}(\rho\Pi_a) \neq \mathrm{tr}(\rho\Pi_b). \tag{62}$$

In that case the classical valuation model merges two outcomes that the quantum statistics separate. If no such preparation is included, the nonseparability remains present combinatorially but is not witnessed by the fragment.

Finally, a true-implies-false (TIFS) or nonfullness constraint becomes visible through nonorthogonal overlap. Suppose a and b are nonorthogonal, but the two-valued states force

$$s(a) = 1 \Longrightarrow s(b) = 0. \tag{63}$$

Classically, conditioning on a being true then forces b to be false. Quantum mechanically, however, the eigenstate Π_a gives

$$\mathrm{tr}(\Pi_a \Pi_b) = |\langle a|b\rangle|^2 > 0 \tag{64}$$

whenever a and b are nonorthogonal. Thus, if the fragment contains the preparation Π_a and the effects Π_a, Π_b, the TIFS implication is operationally contradicted by the nonzero transition probability from a to b.

The important qualification is that these implications are fragment-dependent. A meager hypergraph does not automatically fail every simplex test. If a nonunital atom is never prepared or measured, if a nonseparated pair is never statistically distinguished, or if the endpoints of a TIFS gadget are absent from the preparation/effect list, then the corresponding combinatorial defect need not appear in the operational probability table. Conversely, once the relevant eigenstate preparations and effects are included, the valuation defect becomes an operational obstruction, and the noisy simplex linear program quantifies how much depolarizing noise is needed to hide it.

X. RELATION TO GENERALIZED NONCONTEXTUALITY

The deterministic two-valued-state analysis is closest to the traditional KS setting. Generalized noncontextuality is broader: it does not assume determinism from the outset. Nevertheless, determinism for sharp projective measurements is recovered on the support of eigenstate preparations when the operational equivalences include the relevant projective measurements and mixture decompositions. This is the familiar route from operational noncontextuality to KS-style constraints, developed in modern form by Kunjwal and Spekkens [48].

For a fixed accessible GPT fragment with all operational equivalences already quotiented out, the factorization Eq. (27) is the corresponding cone form of the simplex embedding problem. The computations in this paper choose a smaller fragment, so the resulting r-values are robustnesses of that fragment, not automatic lower or upper bounds on a separately specified full operational robustness. The closure computation in Table II is one intermediate audit: it restores the unit, residual sharp-measurement outcomes, complements, coarse grainings, and some preparation equivalences generated by the same vector data. It is still not a claim to enumerate every possible laboratory realization of the scenario.

In this situation the two-valued-state polytope $\mathsf{P}_2(H)$ is the deterministic shadow of the operational simplex model. Meagerness of $\mathcal{S}_2(H)$ supplies faces or equalities that the Born vectors may violate. Depolarizing noise moves the Born vectors toward a more symmetric point, and the noisy simplex linear program computes the amount of noise needed to make the entire fragment simplex-embeddable.

The crucial qualification is that the simplex linear program is not merely "the fractional coloring polytope." Its variables are the entries of σ in Eq. (27), indexed by facets of the state and effect cones. The number of such facets need not equal the number of vertices or contexts of the hypergraph. In the Yu–Oh 13-ray computation below, there are 13 projectors but 24 facets of each cone, hence σ is 24×24.

XI. COMBINATORIAL CONTEXTUALITY: CHROMATIC COMPLETENESS

Chromatic contextuality imposes a different global requirement. Let $H = (V, \mathcal{C})$ be d-uniform. A proper d-coloring is a map

$$c : V \to \{1, \dots, d\} \tag{65}$$

such that every context contains every color exactly once. Equivalently, for every color k define

$$s_k(v) = \begin{cases} 1, & c(v) = k, \\ 0, & c(v) \neq k. \end{cases} \tag{66}$$

Then each s_k is a two-valued state and

$$\sum_{k=1}^{d} s_k(v) = 1 \qquad \text{for every } v \in V. \tag{67}$$

Thus chromatic colorability asks not merely for two-valued states, but for a complete decomposition of the unit assignment into d admissible two-valued states. The obstruction $\chi(H) > d$ means that no such global spectral-label decomposition exists.

This explains why chromatic contextuality is independent of ordinary valuation scarcity. A hypergraph may possess a separating or even full set of two-valued states and nevertheless fail Eq. (67). Conversely, a KS hypergraph with no two-valued states is of course chromatically contextual, but the chromatic description is then not the most discriminating one: the stronger valuation obstruction is already present.

XII. COMPUTATIONAL PROTOCOL

The computational analysis fixes one convention for all examples. The listed rays define rank-one projectors, these projectors are used both as preparations and as effects, and depolarizing noise is applied only on the state side. The named hypergraphs determine the projector sets and are used in the accompanying combinatorial analysis; in the linear program, their blocks enter only through the operator geometry of the included projectors.

The computations use the non-orthonormal coordinate convention described in Section III. For exact rational audits the program represents real symmetric projectors in the raw coordinate basis

$$(\rho_{11}, \ldots, \rho_{dd}, \rho_{12}, \rho_{13}, \ldots, \rho_{d-1,d}), \tag{68}$$

rather than in a trace-orthonormal basis. In this basis the trace pairing is represented by the diagonal matrix

$$G = \operatorname{diag}\Big(\underbrace{1, \ldots, 1}_{d}, \underbrace{2, \ldots, 2}_{d(d-1)/2}\Big), \tag{69}$$

because the d diagonal entries contribute once whereas the $d(d-1)/2$ off-diagonal real symmetric entries contribute twice to the trace inner product. Thus, if e and ρ denote raw coordinate vectors, then

$$\operatorname{tr}(e\rho) = e^T G\rho. \tag{70}$$

Equivalently, the code may store the dual effect vector Ge, whose ordinary dot product with ρ gives the same Born probability. This convention avoids square-root factors for integer rays and makes the rational cases suitable for exact polyhedral conversion with `pycddlib`/cddlib over $\mathbb{Q}$.

The audit uses the accessible-fragment formalism. Starting from full raw coordinate matrices V_Ω^F and V_E^F, Stage 1 replaces them by accessible coordinate matrices V_Ω^A and V_E^A of full row rank, together with inclusion matrices I_Ω and I_E. These inclusion matrices embed the accessible coordinates back into the full raw coordinate space, so that

$$V_\Omega^F = I_\Omega V_\Omega^A, \qquad V_E^F = I_E V_E^A \tag{71}$$

up to the chosen basis convention. Stage 2 computes facet descriptions H_Ω and H_E of the accessible cones. Stage 3 solves the metric-aware noisy simplex equation

$$I_E^T G\big[(1-r)\,\mathrm{Id}_{\mathcal{A}} + rD\big] I_\Omega = H_E^T \sigma H_\Omega,\ \sigma \geq_{\mathrm{e}} 0,\ 0 \leq r \leq 1. \tag{72}$$

This is the same equation as Eq. (30), written after the accessible reduction and in the raw-coordinate metric convention.

Candidate optima are found numerically. For the rational rows reported as exact optima, the proposed values are then checked symbolically with rational G, H_Ω, H_E, r, and σ, and exact rational dual witnesses are verified.

The certificate status in Table I distinguishes two cases. "Exact optimum" means that both an exact primal certificate and an exact dual certificate have been verified. "Numerical" means that only a floating-point solution of the linear program and residual check are available. The completed pentagon is algebraic rather than rational in the chosen coordinates and is therefore presently treated by the numerical branch. The Specker bug, a truncated form of the KS Γ_1, and Γ_3 of KS [6] are also retained as numerical rows from the earlier high-precision computation.

For the numerical rows, the printed digits are descriptive rather than certified. They should be read together with the residual information in Appendix A. In particular, the algebraic pentagon entry reports the residual of the current numerical branch; the older Specker bug and Γ_3 rows are retained as high-precision numerical evidence until exact algebraic or rationally equivalent certificates are produced.

The symmetric choice used in this audit,

$$\Omega = \mathcal{E} = \{\Pi_v : v \in R\},$$

is not part of the simplex-embeddability criterion itself. The general criterion allows the preparation and effect cones to be different. The present choice is a benchmarking convention: every listed ray is assumed to be both sharply preparable and sharply measurable, so that the Born table contains all projector overlaps $\operatorname{tr}(\Pi_v\Pi_w)$. If instead one used only a single orthonormal basis of preparations, the resulting fragment would usually be simplex-embeddable. Indeed, for preparations $\rho_k = |k\rangle\langle k|$ and arbitrary projective effects $\Pi_v = |v\rangle\langle v|$, the classical model with ontic states $\lambda = k$, preparation distributions $\mu_k(\lambda) = \delta_{\lambda k}$, and response functions $\xi_v(\lambda) = |\langle\lambda|v\rangle|^2$ reproduces

$$\operatorname{tr}(\Pi_v\rho_k) = \sum_\lambda \xi_v(\lambda)\mu_k(\lambda).$$

Thus such a preparation-restricted fragment cannot by itself witness the valuation scarcity of the underlying ray hypergraph.

The second audit uses the same vector coordinates but replaces the projector-only effect set by a vector-generated operational closure. For every mutually orthogonal subset of the listed rays, the program forms the sharp measurement consisting of those rank-one projectors together with the residual effect $\mathbb{1}-\sum_i \Pi_i$, whenever this residual is nonzero. It then adds the unit effect and all nonempty proper coarse grainings, and identifies equal operators in the raw coordinate quotient. Two preparation variants are reported. The *closed-effects* variant keeps the original eigenstate preparations and adds the maximally mixed state. The *closed-effects plus preparations* variant also adds normalized versions of the closed measurement events as preparations whenever their trace is nonzero. These closure rows are computed from exact rational facets when possible, but the values in Table II are numerical unless an exact primal and dual certificate is explicitly stated. For the large Tkadlec closure rows, the linear program was solved in a cutting-dual form: this gives the same numerical optimum but does not produce a primal σ certificate.

XIII. EXAMPLES AND RESULTS

A. Calibration computations: L_{12} and the completed pentagon

The firefly logic L_{12} is included as a calibration for complementarity without simplex obstruction. Its five projectors span a four-dimensional accessible state space inside the six-dimensional real symmetric qutrit trace space. After the Stage 1 accessible-fragment reduction the exact rational audit finds five facets for each accessible cone and returns

$$r_{L_{12}} = 0, \tag{73}$$

with both exact primal and exact dual certificates. Thus L_{12} is not merely a lower-rank exception to the full-rank trace-space code; it is an exactly simplex-embeddable accessible fragment, as expected from its partition-logic realization.

The completed pentagon contains 10 projectors and spans the full six-dimensional real symmetric qutrit trace space. The computation verifies adjacent orthogonality, obtains five contexts of the form Eq. (49), finds 12 cone facets, and returns

$$r_{C_5} \simeq 0.119140568134. \tag{74}$$

The value is numerical only in the present audit because the coordinates involve algebraic quantities outside the rational cddlib backend. It is small compared with the later KS-type examples, as expected: the pentagon is cyclic and already inequality-contextual in the KCBS sense, but it is still far weaker than a KS or nonunital obstruction.

B. Some meager-state configurations

The 13-ray Specker bug [29] is a true-implies-false gadget and hence a nonfull two-valued-state example. The KS Γ_3 configuration has a nonseparating set of two-valued states. Tkadlec's coordinatization supplies a nonunital Schutte-type configuration [44]. These examples realize the three nonempty forms of meagerness: nonfull, nonseparating, and nonunital. In the projector-only simplex test, these particular ray sets yield different robustness values. This shows inequivalence of the chosen fragments, not a universal ordering of the three forms of meagerness.

For the Specker bug, the program uses 13 rays, obtains 26 facets, and returns

$$r_{\text{Specker bug}} \simeq 0.31645570. \tag{75}$$

For Γ_3, the input contains 26 listed rays but 25 unique projectors after identifying antipodal rays. The cone has 230 facets and the linear program returns

$$r_{\Gamma_3} \simeq 0.50219773. \tag{76}$$

For Fig. 2, the 37-ray nonunital example has 276 facets. The numerical optimum rationalizes to

$$r_{\text{nonunital}} = \frac{35}{64}, \tag{77}$$

and the rational audit verifies both exact primal and exact dual certificates at this value. Thus the nonunital row is an exact optimum for the restricted projector-cone fragment. The three Tkadlec-style rows should not be read as universal invariants of the labels "nonfull", "nonseparating", and "nonunital".

C. Yu–Oh: 13 rays versus 25-ray orthogonal completion

The original Yu–Oh (YO) construction consists of the 13 qutrit rays

$$\begin{aligned}
&(1,0,0),(0,1,0),(0,0,1),\\
&(0,1,1),(0,1,-1),(1,0,1),(1,0,-1),(1,1,0),(1,-1,0),\\
&(1,1,1),(1,1,-1),(1,-1,1),(-1,1,1).
\end{aligned} \tag{78}$$

The projector fragment has 24 cone facets. The simplex linear program gives

$$r_{\text{YO13}} = \frac{3}{8}. \tag{79}$$

In the facet ordering returned by the exact audit program, an optimal certificate can be chosen diagonal, with 18 nonzero diagonal entries and exact residual zero.

The 25-ray version used here is the orthogonal completion of the 13 rays: for every orthogonal pair among the original 13 rays, the cross-product ray completing the triad is added, and duplicates are removed projectively. The completion contains 25 projectors, its cone has 96 facets, and the linear program returns

$$r_{\text{YO25}} = \frac{21}{44} \simeq 0.47727273. \tag{80}$$

The larger value is not surprising: the enlarged projector set generates a different operational cone and supplies more Born constraints to be simulated. The rational audit verifies exact primal and exact dual certificates at $r = 21/44$, so this value is an exact optimum for the restricted projector-cone fragment.

The vector-generated closure audit in Table II makes the cost of the restriction explicit. For the 13-ray set, closing the effects while keeping the original eigenpreparations plus the maximally mixed state raises the numerical threshold from 3/8 to 9/20. Closing the preparation side as well gives 21/44, matching the 25-projector completion. Thus the completion row is not just a larger list of rays; in this audit it coincides with the cone geometry obtained after adding the completed rank-one outcomes and their coarse-grained partners.

D. Cabello 18–9 and the Peres–Mermin 24–24 completion

The Cabello–Estebaranz–García-Alcaine 18–9 set is a four-dimensional KS configuration with 18 rays in 9 tetrads [7]. The projector-only fragment is represented in the 10-dimensional real symmetric trace basis. The cone generated by the 18 projectors has 146 facets. The numerical optimum rationalizes to 1/3, and the exact rational audit verifies exact primal and exact dual certificates,

$$r_{18/9} = \frac{1}{3}. \tag{81}$$

so the value is an exact optimum for the restricted projector-cone fragment.

The Peres–Mermin square, analyzed through matrix pencils, yields the 24-24 completion. The 24 vectors include the Cabello 18-ray set as a subset; the remaining rays supply the orthogonal completion associated with the full 24–24 Peres configuration [21]. In the projector-only simplex test, the 24 projectors generate a cone with 120 facets. Writing PM24 for this Peres–Mermin 24-ray fragment, the numerical optimum rationalizes to 4/9, and the exact rational audit verifies exact primal and exact dual certificates,

$$r_{\mathrm{PM24}} = \frac{4}{9}. \tag{82}$$

The larger value is consistent with adding projectors and hence strengthening the restricted fragment. This row is also an exact optimum for the restricted projector-cone fragment.

E. Vector-generated operational closures

Table II reports the second audit, in which the same rational vector sets are closed under the operational data generated by their orthogonality relations. The *closed-effects* rows use the original rank-one eigenpreparations together with the maximally mixed state, and close the effect side under the unit, residual sharp-measurement outcomes, complements, coarse grainings, and equality of operators. The *closed effects+preparations* rows also add normalized versions of the closed effects as preparations. The table does not include the algebraic pentagon or the older Specker-bug and Γ_3 numerical rows, because those were not part of the rational vector-closure script.

The closure rows sharpen the comparison with the projector-cone proxy. Firefly remains simplex-embeddable after closure, as expected for a partition-logical calibration. Yu–Oh 13 increases from 3/8 to 9/20 when the effect closure is added, and to 21/44 when the preparation side is closed as well. The 25-ray Yu–Oh completion is already stable under this closure. Tkadlec's nonunital configuration becomes slightly more robust under closure: 35/64 in the projector cone, 445/792 with closed effects, and 192/337 with closed effects and preparations. Cabello 18–9 moves from 1/3 in the projector cone to 4/9 after effect closure, the same value as the Peres–Mermin 24-ray completion. Peres–Mermin/Peres 24 is stable at 4/9 under the reported closures.

XIV. INTERPRETATION OF THE TABLES

Table I answers one narrow question: how much state-side depolarizing noise is needed before the Born bilinear form on the cones generated by the listed projectors admits a simplex embedding? It does not give chromatic numbers, counts of two-valued states, or full operational robustnesses for complete contextuality scenarios. Table II answers a second, more operational question for the rational vector rows: what changes when the unit, residual sharp-measurement outcomes, complements, coarse grainings, and simple preparation closures generated by the same vectors are added?

Rows marked “exact optimum” are certified by rational primal and dual witnesses. Rows marked “numerical” or “numerical/algebraic branch” remain floating-point evidence rather than independent exact certificates.

The comparisons should therefore be read as diagnostics rather than rankings. L_{12} is a classically representable partition logic and has $r = 0$ even after closure. The completed pentagon gives a small positive numerical value, as expected for the first cyclic probability-level example. The three Tkadlec-style rows show that the chosen nonfull, nonseparating, and nonunital fragments are distinct under this linear program, not that the labels themselves have a universal scalar order. The Yu–Oh and Cabello/Peres–Mermin pairs show that orthogonal or operational completion can change the cone and can raise the threshold. The useful content is the uniform computation together with the certificate status of each row.

TABLE I. Projector-cone depolarizing thresholds for the prepare-and-measure fragments used here. The listed rays are used both as rank-one preparations and rank-one effects, with state-side depolarizing noise $\rho \mapsto \mathrm{tr}(\rho)\mathbb{1}_d/d$. "Exact optimum" means exact primal and dual certificates. "Numerical" means floating-point evidence only.

Configuration	d	projectors	accessible dim.	facets	r	certificate status
Firefly logic L_{12}	3	5	4	5	0	exact optimum
Completed pentagon C_5	3	10	6	12	0.119140568134	numerical/algebraic branch
Specker bug/TIFS, nonfull	3	13	6	26	0.31645570	numerical
Γ_3, nonseparating	3	25	6	230	0.50219773	numerical
Tkadlec Fig. 2, nonunital	3	37	6	276	35/64	exact optimum
Yu–Oh 13-ray projector fragment	3	13	6	24	3/8	exact optimum
Yu–Oh 25-ray orthogonal completion	3	25	6	96	21/44	exact optimum
Cabello–Estebaranz–García-Alcaine 18–9	4	18	10	146	1/3	exact optimum
Peres–Mermin 24–24 completion	4	24	10	120	4/9	exact optimum

TABLE II. Vector-generated operational-closure thresholds. Here $(|\Omega|, |E|)$ gives the number of preparation and effect generators after operator equality has been quotiented, and (f_Ω, f_E) gives the numbers of state- and effect-cone facets. The entries are numerical outputs of the closure script; fractions are rationalizations of the floating-point optima unless exact certificates are separately supplied.

Configuration	closure variant	d	$(\lvert\Omega\rvert, \lvert E\rvert)$	(f_Ω, f_E)	solver	r
Firefly logic L_{12}	closed effects	3	$(6, 11)$	$(5, 5)$	primal	0
Firefly logic L_{12}	closed effects+preparations	3	$(11, 11)$	$(5, 5)$	primal	0
Yu–Oh 13	closed effects	3	$(14, 51)$	$(24, 96)$	primal	9/20
Yu–Oh 13	closed effects+preparations	3	$(51, 51)$	$(96, 96)$	primal	21/44
Yu–Oh 25 completion	closed effects	3	$(26, 51)$	$(96, 96)$	primal	21/44
Yu–Oh 25 completion	closed effects+preparations	3	$(51, 51)$	$(96, 96)$	primal	21/44
Tkadlec Fig. 2, nonunital	closed effects	3	$(38, 123)$	$(276, 1140)$	cutting dual	445/792
Tkadlec Fig. 2, nonunital	closed effects+preparations	3	$(123, 123)$	$(1140, 1140)$	cutting dual	192/337
Cabello 18–9	closed effects	4	$(19, 121)$	$(146, 120)$	primal	4/9
Cabello 18–9	closed effects+preparations	4	$(121, 121)$	$(120, 120)$	primal	4/9
Peres–Mermin/Peres 24	closed effects	4	$(25, 139)$	$(120, 120)$	primal	4/9
Peres–Mermin/Peres 24	closed effects+preparations	4	$(139, 139)$	$(120, 120)$	primal	4/9

XV. DISCUSSION

The examples support a modest conclusion. Valuation tests, simplex-embedding tests, and product-rule tests often agree on canonical KS configurations, but they need not expose the same obstruction after the data have been restricted.

This is clearest for operator-valued parity proofs. If an argument is fully resolved into spectral projectors, it may become an ordinary projective quantum-logic problem, as in the Peres–Mermin transcription to the 24-ray completion. If the same argument is kept at the level of coarse-grained commuting observables, the relevant extra assumption may instead be multiplicativity of assigned values. GHZ isolates that point: the collective joint eigensystem is one Boolean block, but the attempted decomposition of its global product observables into context-independent local factors gives the wrong product sign.

The computational tables have the same limited status. Table I compares selected projector cones and reports which rational values are actually certified. Table II compares a particular vector-generated operational closure of the rational rows. It is not a universal ranking of contextuality. The Specker bug, Γ_3, and Tkadlec's nonunital configuration realize different kinds of two-valued-state meagerness; Yu–Oh 13 and Yu–Oh 25, and likewise Cabello 18–9 and Peres–Mermin/Peres 24, generate different cones after orthogonal or operational completion.

Partition logics and chromatic colorability sit naturally on the incidence side of this comparison. Partition logics show that complementarity and non-Boolean pasting do not by themselves force nonclassical probabilities. Strong colorability asks for a global decomposition of the unit assignment into spectral labels, a stricter condition than the mere existence of two-valued states. Both diagnostics are useful, but neither is the same object as a simplex robustness computed from a probability table.

XVI. CONCLUSION

The useful distinction is therefore not between three new kinds of physics, but between three choices of data: incidence data for valuation, partition-logic, and coloring questions; convex-operational data for simplex-embedding questions; and operator data for functional-composition and product-rule questions.

With that restriction in view, the two main contributions are limited but concrete: the GHZ discussion isolates multiplicative product preservation as the ingredient that can survive inside a single collective Boolean context, and the simplex audits give a uniform set of depolarizing thresholds for both projector-cone fragments and vector-generated closures. The rational projector-cone rows in Table I are exact optima. The closure rows in Table II show how much the reported thresholds can change once unit effects, residual sharp outcomes, complements, coarse grainings, and normalized event preparations are restored.

DATA AND CODE AVAILABILITY

The computations in Tables I and II rely on ray coordinates, incidence files, accessible-fragment reductions, rational facet data, primal/dual certificate checks, and the vector-closure script. These data and scripts should be deposited as supplementary material or in a public repository before submission, together with the bibliography database needed to compile the references. Until that deposit is made, the projector-cone rows marked "exact optimum" are reproducible from the rational primal and dual certificates described in Appendix A; closure rows and algebraic/numerical rows should be treated as numerical evidence unless exact certificates are explicitly provided.

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Appendix A: Computational audit data

This appendix records the computational status behind Tables I and II. It is included because the distinction between numerical optima, exact primal certificates, and exact dual certificates is essential for interpreting the tables.

The exact audit uses the accessible-fragment form of Eq. (72). Integer or rational rays are converted to raw real-symmetric projector coordinates, effects are multiplied by the trace metric $G = \mathrm{diag}(\underbrace{1,\ldots,1}_{d},\underbrace{2,\ldots,2}_{d(d-1)/2})$, and cddlib is used over $\mathbb{Q}$ to obtain rational facet matrices whenever possible. A numerical linear program first proposes r and σ. The proposed rational values are then checked exactly by verifying

$$H_E^T \sigma H_\Omega = I_E^T\big[(1-r)I + rD\big]I_\Omega, \qquad \sigma \geq_{\mathrm{e}} 0. \tag{A1}$$

This is a primal feasibility certificate. Exact optimality additionally requires a rational dual witness. The current audit verifies such dual witnesses for all rational rows reported as exact optima in Table I.

The projector-cone Python audit returned the status summarized in Table III. The table is intentionally diagnostic rather than a second robustness table: its purpose is to show which entries are exact optima and which remain numerical. For the rational rows, exact primal feasibility is checked by Eq. (A1) and exact optimality is certified by rational dual witnesses recovered from the numerical dual active set when direct rationalization is insufficient.

The completed pentagon is not exact-certified by the rational backend because the coordinates in Eq. (48) are algebraic. The numerical branch found 10 projectors, full accessible dimension six, 12 facets, numerical robustness $r = 0.119140568134$, maximum residual approximately 1.8×10^{-13}, and 16 active σ entries. The Specker bug and Γ_3 are retained as numerical rows from the earlier high-precision run. Exact certification of these algebraic rows would require either an algebraic-number polyhedral backend or a rationally equivalent reformulation.

The vector-closure audit used a second script. For each rational vector set it generated residual sharp-measurement effects, complements, coarse grainings, and duplicate-operator quotients from the vector data itself. The run reported in Table II used exact cddlib facets, but exact primal/dual certificates were not requested for the primal closure rows. The large Tkadlec closure rows were solved by a cutting-dual linear program; this is numerically much smaller than the primal σ-program but does not return a primal σ certificate. Consequently the fractions in Table II should be read as rationalizations of numerical optima, not as exact-certified robustnesses.

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TABLE III. Audit status of the simplex computations. Here k_Ω and k_E are the accessible state and effect dimensions after Stage-1 reduction. The rational rows marked "exact optimum" have both exact primal and exact dual certificates.

Configuration	projectors	k_Ω	k_E	facets	r	exact primal	exact dual	note
Firefly logic L_{12}	5	4	4	5	0	yes	yes	exact optimum
Yu–Oh 13	13	6	6	24	3/8	yes	yes	exact optimum
Yu–Oh 25 completion	25	6	6	96	21/44	yes	yes	exact optimum
Tkadlec Fig. 2, nonunital	37	6	6	276	35/64	yes	yes	exact optimum
Cabello 18–9	18	10	10	146	1/3	yes	yes	exact optimum
Peres–Mermin/Peres 24	24	10	10	120	4/9	yes	yes	exact optimum
Completed pentagon C_5	10	6	6	12	0.119140568134	no	no	algebraic/numerical branch
Specker bug/TIFS	13	6	6	26	0.31645570	no	no	numerical high-precision run
Γ_3	25	6	6	230	0.50219773	no	no	numerical high-precision run

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